DECK 18 · WEEK 5 · AFTERNOON 2

Options — Payoffs, Strategies & Pricing

From one-line payoffs to a discretized Black-Scholes

CFA Level I · Vol 7 Learning Modules 8–10

Roadmap — Derivatives II: Options — Payoffs, Strategies & Pricing

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Option payoffs & positions

Call/put payoffs · long vs short · covered call · protective put · VFO Biomian hedge examples

02

Value drivers & moneyness

Exercise + time value split · ITM/ATM/OTM · no-arbitrage bounds · six-factor sensitivity table

03

Put-call parity

Protective put = fiduciary call · S + p = c + PV(X) · synthetics · arbitrage trade · forward parity · firm value

04

One-period binomial

u/d factors · hedge ratio · risk-neutral probability · Hightest Capital example · Black-Scholes as limit

Learning objectives

By the end of this deck you should be able to:

1. Draw the payoff diagrams for long/short calls and puts, and for the two most-traded overlays: the covered call and the protective put.

2. Decompose any option premium into exercise value and time value, classify it as ITM / ATM / OTM, and verify it respects the no-arbitrage bounds.

3. State the six factors that move option value and recite the sign of each effect for calls and for puts.

4. Derive put-call parity from a protective-put vs fiduciary-call no-arbitrage argument, exploit a parity violation for a riskless profit, and build a synthetic long stock from the other three positions.

5. Price a European option on a one-period binomial tree via both the delta-hedge approach and the risk-neutral expectation, and explain why the actual probability is irrelevant to the price.

SECTION 01

Option payoffs & positions

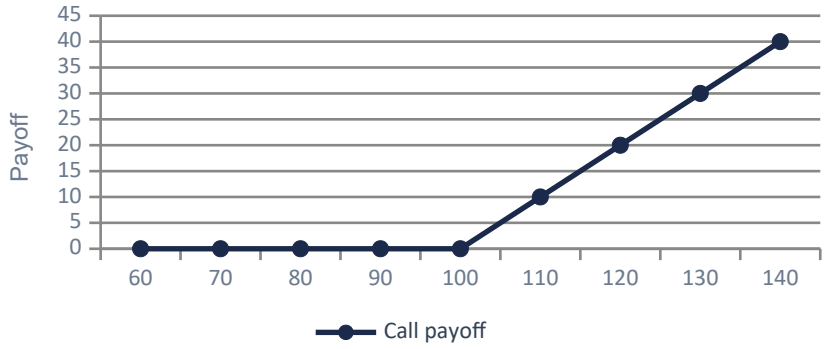
An option's payoff at expiry is Max(0, intrinsic value) — asymmetric, not linear like a forward

The kink at the strike is the entire source of option-market richness: below it the payoff is flat, above it grows one-for-one.

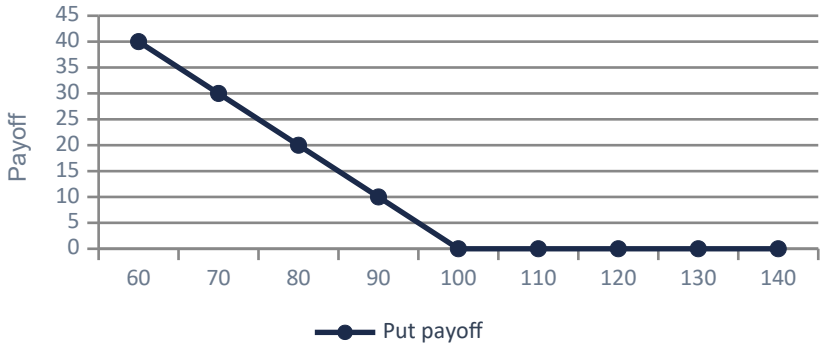
$$cT = Max(0, ST - X) \qquad pT = Max(0, X - ST)$$

where: ST = underlying spot at expiry T; X = exercise (strike) price. The premium paid at t = 0 offsets the payoff to give the profit.

Long call — strike 100



Long put — strike 100



Source: CFA Institute, Curriculum Vol 7, Learning Module 8, Exhibit 3 and pp. 180–182.

Long and short positions are mirror images around the premium — the buyer's max loss equals the seller's max gain

The option writer's P&L is the counterparty of the buyer's: sum of the two profits is always exactly zero, minus transaction costs.

Position	Payoff at T	Max loss	Max gain	Breakeven
Long call	$\text{Max}(0, S_T - X)$	Premium c_0	Unlimited	$X + c_0$
Short call	$-\text{Max}(0, S_T - X)$	Unlimited	Premium c_0	$X + c_0$
Long put	$\text{Max}(0, X - S_T)$	Premium p_0	$X - p_0$ (if $S_T = 0$)	$X - p_0$
Short put	$-\text{Max}(0, X - S_T)$	$X - p_0$	Premium p_0	$X - p_0$

FOUR INTUITIONS TO LOCK IN

- Long options have capped downside (the premium) and, for calls, uncapped upside — you pay for the optionality.
- Short options have capped upside (the premium) and, for calls, uncapped downside — you are selling insurance.
- Short puts are the trade with the worst tail: you can lose $X - p_0$ if the underlying drops to zero.
- The premium is never part of the payoff at T — it's a sunk cost paid at $t = 0$ that shifts profit vertically.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, pp. 180–184.

A covered call sells upside in exchange for option income — the most widely used option overlay in asset management

Long stock + short call: if ST ≤ X, keep the stock and the premium; if ST > X, forfeit upside beyond X but still collect the premium.

Covered-call payoff = ST − Max(0, ST − X) = Min(ST, X)
(plus premium c_0 received at t = 0)

CFA LM 9 EXAMPLE 3 — VISWAN FAMILY OFFICE BIOMIAN COVERED CALL

VFO holds non-dividend Biomian stock at S_0 = INR 295. CIO sells a 6-month call at X = INR 325 to generate income. Risk-free rate 4%; put (same X, T) priced at INR 56.

1. Covered call position = $S_0 - c_0$ (long stock, short call)
2. By put-call parity: $S_0 - c_0 = X(1+r)^{-T} - p_0$
3. Substitute: $295 - c_0 = 325 \times 1.04^{-0.5} - 56 = 318.69 - 56 = 262.69$
4. Solve: $c_0 = 295 - 262.69$ = INR 32.31

Sell 6-month X = 325 call for INR 32.31; total income = premium. Max payoff at T capped at 325 + 32.31 = 357.31.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, Example 3 "VFO Covered Call", pp. 204–206.

A protective put buys insurance against downside at a known upfront cost — the floor everyone wants and few want to pay for

Long stock + long put: if $S_T \geq X$, let the put lapse; if $S_T < X$, exercise and sell at X — a floor at X, paid for with premium p_0.

$$\text{Protective-put payoff} = S_T + \text{Max}(0, X - S_T) = \text{Max}(S_T, X)$$
$$\text{(less premium } p_0 \text{ paid at } t = 0\text{)}$$

CFA LM 9 EXAMPLE 1 — VFO HEDGING BIOMIAN WITH A 6-MONTH PUT AT X = 265

S_0 = INR 295, X = INR 265, observed 6-month call at same X priced at c_0 = INR 59, r = 4%, T = 0.5.

1. Put-call parity: $S_0 + p_0 = c_0 + X(1 + r)^{-T}$
2. $PV(X) = 265 \times 1.04^{-0.5}$ = INR 259.85
3. Solve: $295 + p_0 = 59 + 259.85$
4. $p_0 = 318.85 - 295$ = INR 23.85

Put premium INR 23.85 ⇒ hedged floor at 265; VFO caps downside while retaining upside above 265.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, Example 1 "Biomian Put-Call Parity", pp. 202–203.

A collar stacks a protective put and a covered call on the same stock — floor plus cap, funded cheaply by the call premium

When the call premium collected roughly equals the put premium paid, it's a zero-cost collar — a favourite of founders hedging concentrated equity.

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Collar payoff at T = Max(X_put, Min(ST, X_call)) (+ net premium, usually ≈ 0)
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where: X_put < S_0 < X_call. Downside floored at X_put, upside capped at X_call; profile is a "squashed stock".

ZERO-COST COLLAR — BIOMIAN CEO HEDGING A CONCENTRATED HOLDING

CEO long 100k shares at S_0 = INR 295. Wants to lock in a 10% downside floor and fund it by capping upside. r = 4%, T = 1y.

1. Buy 6-month put at X_put = 265 ⇒pay p_0 = INR 23.85 per share (from parity example)
2. Sell 6-month call at X_call = 325 ⇒receive c_0 = INR 32.31 per share (from covered-call example)
3. Net premium = +32.31 − 23.85 = +INR 8.46 per share INFLOW (collar earns carry)
4. Downside at T: floored at 265. Upside at T: capped at 325. P&L band: −30 / +30 per share before premium.

Net cash-positive collar: floor 265, cap 325, fat INR 8.46/share credit. Works when OTM calls trade rich vs OTM puts.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §3 Option Strategies, pp. 204–206.

SECTION 02

Value drivers & moneyness

An option's price decomposes exactly into exercise value plus time value — the sum must equal the traded premium

Exercise value uses the PV of X (not X itself) — this difference is the first and most common mark-to-market error in practice.

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ct  =  Max(0, St – X(1+r)^(–(T–t)))  +  Time Valuet
        (analogue for puts with X and St swapped)
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where: Time value ≥ 0 always, decays to zero at t = T. A deep-OTM option has price ≈ time value only.

CFA LM 8 SELF-ASSESSMENT Q3 — GBP EUROPEAN CALL DECOMPOSITION

3-month European call on a no-dividend stock. X = GBP 50, r = 2%, St = GBP 57.50. Observed call premium ct = GBP 10.

1. PV of exercise price: $X(1 + r)^{-0.25} = 50 \times 1.02^{-0.25}$ = GBP 49.75
2. Exercise value = Max(0, St − PV(X)) = Max(0, 57.50 − 49.75) = GBP 7.75
3. Time value = ct − exercise value = 10.00 − 7.75 = GBP 2.25
4. Check: 7.75 + 2.25 = 10.00 ✓

Exercise value GBP 7.75 + time value GBP 2.25 = traded premium GBP 10.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §3 & §5 and Self-Assessment Q3, pp. 175–180.

Moneyness classifies options as ITM, ATM, or OTM — and drives how sensitive the option is to the underlying

Delta (∂c/∂S or ∂p/∂S) ≈ likelihood of ending in the money. Deep-ITM calls track the stock one-for-one; deep-OTM options barely move.

Moneyness	Call condition	Put condition	Delta intuition	Typical use
In-the-money (ITM)	St > X	St < X	Approaches ±1 when deep	Substitute for the underlying
At-the-money (ATM)	St = X	St = X	≈ ±0.5	Most sensitive to volatility
Out-of-the-money (OTM)	St < X	St > X	Approaches 0 when deep	Cheap tail-risk

ITM CALLS BEHAVE LIKE STOCK

A call 30% in-the-money with a week to expiry has delta near 1. A 10c move in the stock ⇒~10c move in the call. Useful for leveraged long exposure at controlled cost.

OTM OPTIONS ARE ALL TIME VALUE

A call 30% out-of-the-money with a week to expiry has delta ≈ 0.02. It's a lottery ticket — cheap per share, but the expected payoff is tiny. Most OTM options expire worthless.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §4 "Option Moneyness", pp. 176–178.

No-arbitrage bounds pin option prices: between Max(0, S − PV(X)) and S for a call, between Max(0, PV(X) − S) and X for a put

Price a call above S and I buy the stock / sell the call for free money. Price a call below S − PV(X) and I buy the call / short-synthetic-the-stock for free money.

$$\text{Max}(0, S_t - PV(X)) \leq c_t \leq S_t \qquad \text{Max}(0, PV(X) - S_t) \leq p_t \leq X$$

CFA LM 8 EXAMPLE 4 — ONE-YEAR EUR CALL BOUNDS AT T − t = 0.5

Exercise price X = EUR 1,000. Six-month forward spot St = EUR 1,050. Risk-free rate r = 1%.

1. PV(X) over remaining 0.5y: $1{,}000 \times 1.01^{-0.5}$ = EUR 995.04
2. Lower bound: Max(0, 1,050 − 995.04) = EUR 54.96
3. Upper bound: St = EUR 1,050
4. Any traded call price outside [54.96, 1,050] offers a riskless arbitrage.

54.96 ≤ ct ≤ 1,050. At-the-money → lower bound small; deep-ITM → lower bound dominates.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §6 "Arbitrage", Example 4, pp. 180–184.

Six factors drive option value; all six signs are known, and volatility is the only factor whose sign is identical for calls and puts

Memorize this table once. The CFA L1 exam tests a sign-flip from it every single sitting — typically risk-free rate or income on underlying.

Factor (increase in ...)	Call value	Put value	One-line intuition
Underlying price (S)	+	–	Calls gain, puts lose as S rises
Exercise price (X)	–	+	Calls are cheaper to exercise when X is low
Time to expiration (T)	+	+/–	More time → wider distribution; put exception[1]
Risk-free rate (r)	+	–	Higher r ⇒ lower PV(X) ⇒ calls up, puts down
Volatility (σ)	+	+	Both sides: asymmetric payoff likes wide tails
Income on underlying	–	+	Div/yield accrues to holder, not to call[2]

[1] Longer time to expiration usually increases put value, but can lower it for deep-ITM puts because the payoff X − ST is discounted.

[2] Income (dividends, coupons, convenience yield) lowers call value because it accrues to the holder of the underlying, not the option.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §8 "Factors Affecting Option Value", Exhibit 9, pp. 186–190.

SECTION 03

Put-call parity

A protective put and a fiduciary call share every future cash flow — so no-arbitrage forces $S_0 + p_0 = c_0 + X(1+r)^{-T}$

Two portfolios that pay the same thing in every state of the world must cost the same today — otherwise, arbitrage.

Protective put: $S_0 + p_0 \equiv c_0 + X(1 + r)^{-T}$ Fiduciary call:

where: European options, same X, same T, no dividends on the underlying, one risk-free rate r.

Portfolio @ t = T	ST < X	ST = X	ST > X
Protective put: S + p	ST + (X − ST) = X	ST = X	ST + 0 = ST
Fiduciary call: c + PV(X)→X	0 + X = X	X	ST − X + X = ST
Cash-flow difference	0	0	0
Therefore prices must match at t = 0	**S + p = c + PV(X)**	**(parity)**	✓

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §2 "Put-Call Parity", Exhibits 1–4, pp. 200–203.

CFA LM 9 Example 1 prices a put directly off a traded call — parity is a pricing engine, not just an identity

Whenever a liquid call and illiquid put exist on the same name and strike, parity gives the put's mark — instantly.

$$p_0 = c_0 + X(1 + r)^{-T} - S_0$$

BIOMIAN 6-MONTH PUT FROM TRADED CALL

Biomian shares S_0 = INR 295. Observed 6-month European call at X = INR 265 trades at c_0 = INR 59. r = 4%, T = 0.5. No dividends.

1. Compute PV of strike: $X(1 + r)^{-T} = 265 \times 1.04^{-0.5} = 265 / 1.0198 =$ INR 259.85
2. Substitute into rearranged parity: $p_0 = c_0 + PV(X) - S_0$
3. $p_0 = 59 + 259.85 - 295$
4. $p_0 = 318.85 - 295 =$ INR 23.85
5. Cross-check: $S_0 + p_0 = 295 + 23.85 = 318.85 = c_0 + PV(X) = 59 + 259.85 = 318.85$ ✓

No-arbitrage 6-month put premium = INR 23.85.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, Example 1 "Biomian Put-Call Parity", pp. 202–203.

CFA LM 9 Example 2 locks in a riskless INR 6.15 per share when the traded put is too rich vs parity

If S + p > c + PV(X), the expensive side is sold and the cheap side is bought. Cash flows net to zero at T; the initial surplus is pure profit.

Leg @ t = 0	Cash flow @ t = 0	ST < X	ST = X	ST > X
Sell stock at S_0	+295	−ST	−ST (=−X)	−ST
Sell put at p_0 (traded, overpriced)	+30	−(X−ST) = ST−X	0	0
Buy call at c_0	−59	0	0	ST − X
Buy risk-free bond to receive X @ T	−259.85	+X	+X	+X
NET	**+6.15**	**0**	**0**	**0**

READING THE GRID

At t = 0 the investor pockets INR 6.15 in cash. At t = T, in every possible ending price ST, the four legs net to exactly zero — there's no residual exposure. This is the textbook definition of a riskless arbitrage: positive cash today, no risk tomorrow.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, Example 2 "VFO Parity Arbitrage", pp. 203–204.

Parity rearranges into a synthetic-position cookbook — each position is replicable from the other three

Read the rows: to synthesize the position on the left, build the combination on the right.

Target position	Underlying	Risk-free bond	Call	Put
Long underlying (S_0)	–	Long X·PV	Long	Short
Long risk-free bond (PV(X))	Long	–	Short	Long
Long call (c_0)	Long	Short X·PV	–	Long
Long put (p_0)	Short	Long X·PV	Long	–

WHY THIS MATTERS IN PRACTICE

Dealers synthesize stock exposure from options daily — because the option market can be deeper and cheaper to finance than the cash market. An equity-like payoff built from call − put + bond is the single most common use of put-call parity in sell-side flow desks.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §3 Exhibit 6 "Replication of Positions", pp. 204–205.

A synthetic long stock is +call − put + cash — identical payoff to owning the underlying outright

Useful when direct share purchase is expensive (dividends to miss, short-borrow to avoid, regulatory capital). The replicating portfolio bypasses the cash market entirely.

$$S_0 = c_0 - p_0 + X(1 + r)^{-T}$$

SYNTHETIC LONG STOCK — BIOMIAN NUMBERS

Same inputs as parity example: 6-month options at X = INR 265, c_0 = 59, p_0 = 23.85, r = 4%, T = 0.5, S_0 = INR 295.

1. Buy call: −INR 59
2. Sell put: +INR 23.85
3. Buy zero-coupon bond maturing at X = 265: −PV(X) = −INR 259.85
4. Total initial outlay: 59 − 23.85 + 259.85 = INR 295 ⇒ equal to buying stock
5. At T: if ST > 265, call pays ST − 265 + bond 265 = ST; if ST < 265, assigned on put: pay ST − 265 + bond 265 = ST. Payoff = ST either way.

Synthetic long stock costs exactly S_0 = 295 today and pays ST at expiry — indistinguishable from the cash stock.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §3 Synthetic Positions, pp. 204–206.

Put-call forward parity replaces S_0 with PV of the forward price — useful when the cash asset is hard to hold but the forward is liquid

For commodities, currencies, or shares with complex carry, traders often start pricing options from the forward, not the spot.

$$F_0(T)\cdot(1+r)^{-T} + p_0 = c_0 + X(1+r)^{-T} \Rightarrow p_0 - c_0 = [X - F_0(T)]\cdot(1+r)^{-T}$$

where: $F_0(T)$ = no-arbitrage forward price. When $F_0(T) = X$, $p_0 = c_0$ — the famous at-the-money-forward equivalence.

CFA LM 9 EXAMPLE 4 — BIOMIAN FROM FORWARD + BOND

$S_0 = 295$, $r = 4\%$, $T = 0.5$ ⇒ forward $F_0(T) = 295 \times 1.04^{0.5}$ = INR 300.84. Same $X = 265$, $c_0 = 59$.

1. Forward-parity form: $p_0 - c_0 = (X - F_0(T)) \cdot (1 + r)^{-T}$
2. $p_0 - 59 = (265 - 300.84) \times 1.04^{-0.5} = -35.84 / 1.0198$
3. $p_0 - 59 = -35.14 \Rightarrow p_0$ = INR 23.86

Same INR 23.86 answer — confirms forward and spot parities are algebraically equivalent.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §5 Example 4 "VFO Put-Call Forward Parity", pp. 207–209.

Put-call parity prices claims on a levered firm: equity is a long call on assets, risky debt is a risk-free bond minus a short put

Merton (1974): treat the firm as assets V, a zero-coupon debt face D, and equity as the residual — the option analogy falls out.

EQUITYHOLDERS

Payoff at T: Max(0, VT − D)

= Long call on firm value with strike D

- Unlimited upside (residual claim)
- Downside limited to zero (limited liability)
- Value-creating projects & vol both boost equity — textbook risk-shifting incentive

DEBTHOLDERS

Payoff at T: Min(VT, D)

= Risk-free bond of face D − Short put on VT with strike D

- Upside capped at D (face amount)
- Downside when VT < D (default)
- Credit spread = price of the sold put — and grows with firm volatility

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §6 "Firm Value Application", pp. 209–212.

SECTION 04

One-period binomial

Pricing a contingent claim requires a model of future prices — the binomial tree gives the simplest honest one

Forwards need no model of the future (payoff is linear). Options do (payoff has a kink) — the binomial tree is the MVP of that model class.

$$S_u = R_u \cdot S_0 \qquad S_d = R_d \cdot S_0 \qquad (R_u > 1 > R_d;$$
$$\text{spread } R_u - R_d = \text{volatility proxy})$$

where: q = real-world probability of an up move; 1 − q = probability of a down move. q turns out to be irrelevant for the price.

TWO OUTCOMES

Over one period, the underlying goes from S_0 to either S_u or S_d. No third state. That's it — everything else follows from arbitrage.

VOLATILITY = SPREAD

The gap between S_u and S_d is the model's view of volatility. Doubling the spread doubles range of possible payoffs ⇒option value rises.

MULTI-PERIOD → TREE

Chain N one-period steps ⇒N-period binomial tree with N+1 terminal nodes. As N → ∞, approaches the Black-Scholes continuous-time model.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §2–3 "The Binomial Model", pp. 221–223.

A portfolio of h shares minus one call can be made risk-free — h is the option's payoff range divided by the asset's

The hedge ratio is the binomial-world delta: the exact share count that makes a short-call / long-stock portfolio invariant to which branch the tree takes.

$$h^* = (c_u - c_d) / (S_u - S_d) \quad \Rightarrow \quad V_u = h^* \cdot S_u - c_u = V_d = h^* \cdot S_d - c_d$$

where: Choose h so portfolio value is identical in both states ⇒portfolio earns r ⇒* $c_0 = h^* \cdot S_0 - V_1/(1 + r)$.

THE THREE-STEP RECIPE

1. Tree the underlying: compute S_u and S_d from R_u, R_d.
2. Tree the option payoffs: $c_u = \text{Max}(0, S_u - X)$, $c_d = \text{Max}(0, S_d - X)$.
3. Compute h* from the payoff ratio, then $V_1 = h^* \cdot S_u - c_u$ (or $- c_d$, they're equal by construction).
4. Discount V_1 at r to get V_0; solve $c_0 = h^* \cdot S_0 - V_0$.

➤ No probabilities appear anywhere. Just R_u, R_d, r and the option payoff.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §4 "Pricing a European Call Option", pp. 223–226.

CFA LM 10 §4 canonical example: $S_0 = 80$, $X = 100$, up to 110 / down to 60, $r = 5\%$ — delta-hedge gives $c_0 = 4.57$

Numbers chosen deliberately: option is OTM today ($S_0 < X$) but ITM in the up state. Walks through every step of the recipe.

ONE-YEAR EUROPEAN CALL · DELTA-HEDGE METHOD

$S_0 = 80$, $X = 100$, $S_u = 110$ ($R_u = 1.375$), $S_d = 60$ ($R_d = 0.75$), $r = 5\%$, $T = 1$.

1. Option payoffs: $c_u = \text{Max}(0, 110 - 100) = 10$, $c_d = \text{Max}(0, 60 - 100) = 0$
2. Hedge ratio $h^* = (c_u - c_d)/(S_u - S_d) = (10 - 0)/(110 - 60) = 10/50 = 0.20$
3. Portfolio value at T (both branches): $V_u = 0.20 \cdot 110 - 10 = 12$; $V_d = 0.20 \cdot 60 - 0 = 12$. ✓Equal by construction.
4. PV the certain $V_1 = 12$: $V_0 = 12 / 1.05 = 11.43$
5. Call price: $c_0 = h^* \cdot S_0 - V_0 = 0.20 \cdot 80 - 11.43 = 16.00 - 11.43 = 4.57$

Fair 1-year call price = 4.57. Check: short 1 call, buy 0.20 stock ⇒ invests 11.43, grows to 12 ⇒ earns exactly 5% ✓

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §4 Exhibit 2–3 worked case, pp. 223–226.

The same price falls out of a risk-neutral expectation — π acts as if investors were risk-neutral, even though they're not

π is a computation tool, not a real probability. It makes the discounted underlying a martingale — the mathematical magic trick of derivatives pricing.

$$\pi = (1 + r - R_d) / (R_u - R_d) \qquad c_0 = [\pi \cdot c_u + (1 - \pi) \cdot c_d] / (1 + r)$$

where: π = risk-neutral probability of an up move. Always 0 < π < 1 if r is between R_d − 1 and R_u − 1.

SAME NUMBERS — RISK-NEUTRAL METHOD

Reuse the canonical example: $S_0 = 80$, $X = 100$, $R_u = 1.375$, $R_d = 0.75$, $r = 5\%$, $c_u = 10$, $c_d = 0$.

1. $\pi = (1 + 0.05 - 0.75) / (1.375 - 0.75) = 0.30 / 0.625 = 0.48$
2. $1 - \pi = 0.52$
3. Expected payoff $= 0.48 \cdot 10 + 0.52 \cdot 0 = 4.80$
4. Discount: $c_0 = 4.80 / 1.05 = 4.57$
5. Matches the delta-hedge answer exactly — as it must, both follow from the same no-arbitrage condition.

$c_0 = 4.57$ ✓ Same number, faster route. Risk-neutral pricing scales directly to N-period trees & Black-Scholes.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §5 "Risk Neutrality", pp. 230–233.

Actual probabilities do not affect the price — only volatility and the risk-free rate do

The CIO's bullish view is irrelevant to the premium. Doubling the move width (from ±20% to ±40%) roughly doubles the call premium.

Scenario	S_u	S_d	R_u	R_d	h*	π	c_0
±20% move (S_0=50, X=55, r=5%)	60	40	1.20	0.80	0.25	0.625	USD 2.98
±40% move (same)	70	30	1.40	0.60	0.375	0.5625	USD 8.04
±20% + bullish view (q=0.80)	60	40	1.20	0.80	0.25	0.625	USD 2.98
±20% + bearish view (q=0.20)	60	40	1.20	0.80	0.25	0.625	USD 2.98

READING THE TABLE

Rows 1→2: doubling volatility lifts call price 2.7x (2.98 → 8.04). Rows 1, 3, 4: changing the real-world probability q from 50% to 80% to 20% leaves the price unchanged at 2.98 — because q does not appear anywhere in the pricing formula. This is the single most counter-intuitive fact in options pricing.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §4 Example 1 "Hightest Capital", pp. 226–228.

As the number of steps grows, the binomial model converges to Black-Scholes — the continuous-time limit that prices most traded options

Pick $R_u = e^{(\sigma\sqrt{\Delta t})}$ and $R_d = e^{(-\sigma\sqrt{\Delta t})}$ at each step, let $\Delta t \to 0$, and the N-period tree's price converges to the Black-Scholes formula.

$$c_0 = S_0 \cdot N(d_1) - X \cdot e^{(-rT)} \cdot N(d_2) \qquad d_{1,2} = [\ \ln(S_0/X) + (r \pm \sigma^2/2)\cdot T\ ] / (\sigma\cdot\sqrt{T})$$

where: $N(\cdot)$ = standard normal CDF; σ = annualized volatility of continuously compounded returns. Continuous analogue of π.

WHY BINOMIAL FIRST

Trees are pedagogy: every step is a replication / arbitrage argument in full view. Black-Scholes is the same argument taken to the continuous-time limit — harder to see but cheaper to compute.

WHERE TREES WIN

American options, Bermudan exercise, path-dependent payoffs, dividends at discrete dates — these need a tree (or Monte Carlo). Plain European options get Black-Scholes; everything else gets a tree.

[1] CFA L1 formula sheet only requires the one-period binomial. Black-Scholes is introduced conceptually here and priced in L2.

[2] Trees remain the industry standard for pricing American options and any feature with early exercise — Black-Scholes doesn't admit early-exercise directly.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §5 closing discussion; LM 8 volatility factor, pp. 232–234.

KEY TAKEAWAYS

Deck 18 — Derivatives II: Options — Payoffs, Strategies & Pricing

- Option payoffs at expiry: $c_T = \text{Max}(0, S_T - X)$, $p_T = \text{Max}(0, X - S_T)$ — asymmetric, kinked, the source of everything that follows.
- Six-factor sensitivity memorised: S, X, T, r, σ, income. Volatility is the only factor with the same sign for calls and puts.
- Synthetics: long stock = +call − put + cash at PV(X); covered call = $S_0 - c_0$ = long bond − short put; protective put = long call + long bond.
- Risk-neutral probability $\pi = (1+r - R_d)/(R_u - R_d)$ gives the same price in one expected-value step; actual probability q is irrelevant.
- Every option price = exercise value + time value; exercise uses PV(X) not X; time value always ≥ 0 and decays to zero at T.
- Put-call parity: $S_0 + p_0 = c_0 + X(1+r)^{-T}$ — derives from a two-portfolio arbitrage, prices any of the four positions from the other three.
- Binomial hedge ratio $h^* = (c_u - c_d)/(S_u - S_d)$ builds a risk-free portfolio; its discounted cost pins the option price.
- Canonical answer (CFA LM10): S_0=80, X=100, up 37.5% / down 25%, r=5% ⇒c_0 = 4.57. Black-Scholes is the continuous-time limit of this tree.